

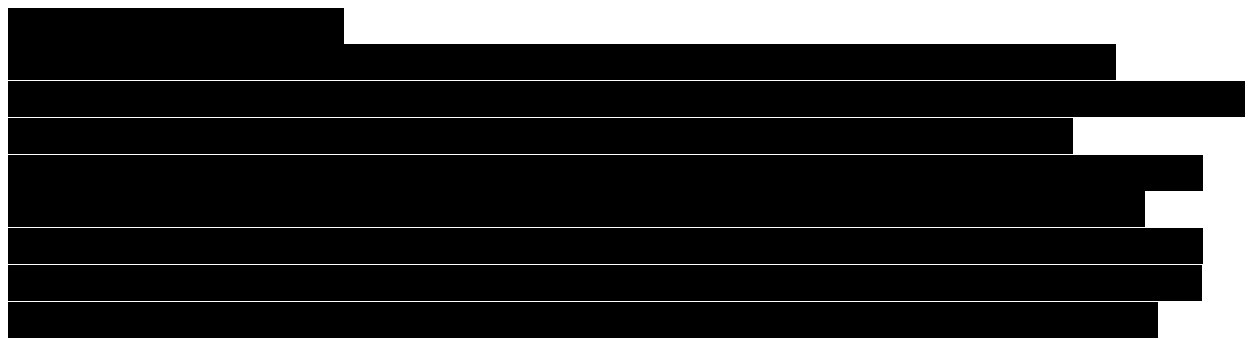
Statement of Work (SOW)

NASA Goddard Space Flight Center (GSFC) Earth Sciences Division

Earthmover Cloud-Native Data Platform Pilot for MERRA-21C

Background:

The purpose of this SOW is to describe the firm fixed price work to be performed by the Contractor to deploy a cloud-native data platform for NASA's Global Modeling and Assimilation Office (GMAO). This work supports NASA's strategic shift toward Open Science by enabling efficient, scalable access to large climate datasets such as MERRA-21C. The pilot will demonstrate Earthmover's ability to convert legacy file-based archives into Analysis-Ready Cloud Optimized (ARCO) data services, reducing barriers for scientific workflows and AI/ML applications.



Scope of Work:

The Contractor shall perform the firm fixed price work necessary to ingest, convert, and serve a subset of the MERRA-21C dataset as ARCO data. The Contractor shall provide qualified personnel, materials, equipment, documentation, and facilities (unless otherwise stated) to perform the work.

Period of Performance:

Start: Upon award

Duration: 3 months

Key milestones will be determined in consultation with Science POC

Project Specific Requirements:

The Contractor shall perform the following:

- **Track 1: ARCO Ingestion & Conversion**
 - Ingest a high-value subset of MERRA-21C (e.g., hourly 3D meteorological fields) from NASA's NCCS Discover HPC.
 - Convert data from NetCDF4 to Icechunk-backed Zarr stores within Earthmover's Arraylake platform.
 - Deliver a "Golden Copy" of cloud-optimized MERRA-21C supporting efficient parallel reads.
- **Track 2: Dynamic Regridding & Aggregation Service**
 - Develop and benchmark dynamic regridding (cubed-sphere to lat-lon) and aggregation capabilities (spatial/temporal/vertical).

- Demonstrate >10x reduction in I/O latency compared to standard NetCDF-over-S3 workflows.
- Provide cost analysis of dynamic regridding vs. precomputed lat-lon storage.
- **Track 3: Public Access & Delivery**
 - Configure Flux API to serve ARCO MERRA-21C data via a public-facing endpoint.
 - Demonstrate zero-download access through REST API and Python client without AWS credentials.
- Provide NASA with compliance documentation for cloud security and data integrity standards.
- Participate in technical reviews and provide performance benchmarking reports.
- Deliver a final white paper co-authored with NASA detailing architecture and performance gains.

Deliverables:

- ARCO version of selected MERRA-21C subset hosted on Arraylake.
- Flux API endpoint for public data access.
- Source code and associated documentation for creating the ARCO version of MERRA-21C
- Benchmarking report (and associated source code) comparing Earthmover pipeline vs. legacy workflows
- Final white paper summarizing pilot results and recommendations

Final Closeout:

This SOW will be considered complete when all deliverables have been provided to NASA and the pilot engagement concludes.

